

SIMATS ENGINEERING

ASSESSMENT TOOL 2 - MEDIUM (Assignment)

COURSE CODE: CSA09

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications. (BL3,BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 3

Total Marks: 30

ANNEXURE A

Assignment

Code of Conduct:

*I **J KARTHIK / 192472136** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: J KARTHIK

ANNEXURE

Assignment

1. Online Shopping Coupon Verification

"An e-commerce website offers the coupon code SAVE50. Write a Java program that:

Removes extra spaces using trim()

Converts the entered code to uppercase using toUpperCase()

Compares it with the original coupon using equalsIgnoreCase()

Q1 Online Shopping Coupon Verification

10 marks

An e-commerce website offers the coupon code SAVE50. Write a Java program that:

Removes extra spaces using trim()

Converts the entered code to uppercase using toUpperCase()

Compares it with the original coupon using equalsIgnoreCase()

Your Code

```
1 import java.util.Scanner;
2 public class CouponVerification {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String originalCoupon = "SAVE50";
6         System.out.print("Enter coupon code: ");
7         String enteredCoupon = sc.nextLine();
8         enteredCoupon = enteredCoupon.trim();
9         enteredCoupon = enteredCoupon.toUpperCase();
10        if (enteredCoupon.equalsIgnoreCase(originalCoupon)) {
11            System.out.println("Coupon is valid.");
12        } else {
13            System.out.println("Coupon is invalid.");
14        }
15        sc.close();
16    }
17 }
```

Input

SAVE50

Output

Enter coupon code: Coupon is valid.

2. Restaurant Digital Order Analyzer

"A restaurant stores orders as DOSA-3-IDLY-2-VADA-1. Create a program that:

Splits the order using split()

Converts food names to uppercase using toUpperCase()

Counts the number of ordered items using length()

Q2 Restaurant Digital Order Analyzer

10 marks

A restaurant stores orders as DOSA-3-IDLY-2-VADA-1. Create a program that:

Splits the order using split()

Converts food names to uppercase using toUpperCase()

Counts the number of ordered items using length()

Your Code

```
1 import java.util.Scanner;  
2 public class Main {  
3     public static void main(String[] args) {  
4         Scanner sc = new Scanner(System.in);  
5         String order = sc.nextLine();  
6         String[] items = order.split(",");  
7         for (String item : items) {  
8             String[] data = item.split("-");  
9             System.out.println(data[0].toUpperCase() + " : " + data[1]);  
10        }  
11        System.out.println("Total Items: " + items.length);  
12    }  
13 }
```

Input

Dosa - 3, Idly - 2, Vada - 1

Output

```
DOSA : 3  
IDLY : 2  
VADA : 1  
Total Items: 3
```

3. Bank Token Queue Management System

"A bank serves customers based on token order. Using Queue:
Add customer tokens using offer()
Serve the next customer using poll()
Display the upcoming customer using peek()"

Q3 Bank Token Management System

10 marks

Customers receive tokens in the order they arrive at a bank counter.

Requirements:

Add customer tokens to a Queue

Serve customers in FIFO order

Display the next customer to be served and the remaining queue

Your Code

```
1 import java.util.*;
2 public class BankQueue {
3     public static void main(String[] args) {
4         Queue<String> customers = new LinkedList<>();
5         customers.add("Token 101");
6         customers.add("Token 102");
7         customers.add("Token 103");
8         customers.add("Token 104");
9         customers.add("Token 105");
10        System.out.println("Customer Queue: " + customers);
11        System.out.println("Next customer to be served: " + customers.peek());
12        System.out.println("\nServing customers:");
13        while (!customers.isEmpty()) {
14            String customer = customers.poll();
15            System.out.println("Serving: " + customer);
16            System.out.println("Remaining queue: " + customers);
17        }
18    }
19 }
```

Run

Save

Input

stdin input

Output

```
Customer Queue: [Token 101, Token
102, Token 103, Token 104, Token
105]
Next customer to be served: Token
101
```

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification